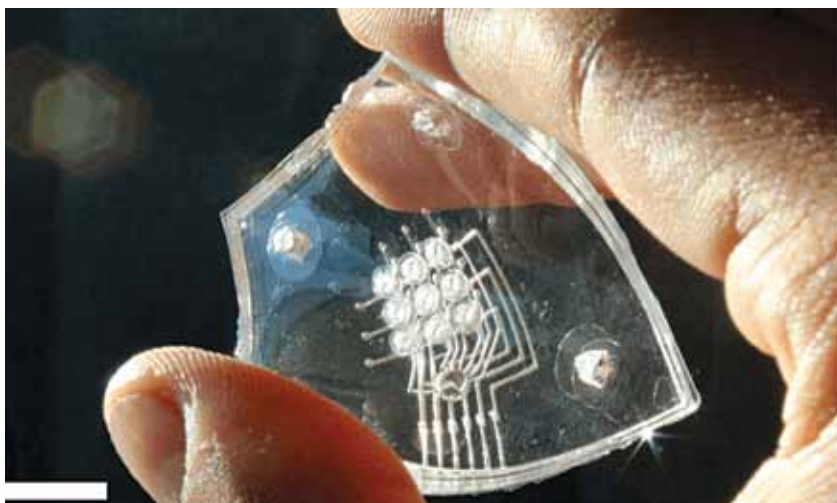




Braille 'display' the visually-impaired can feel

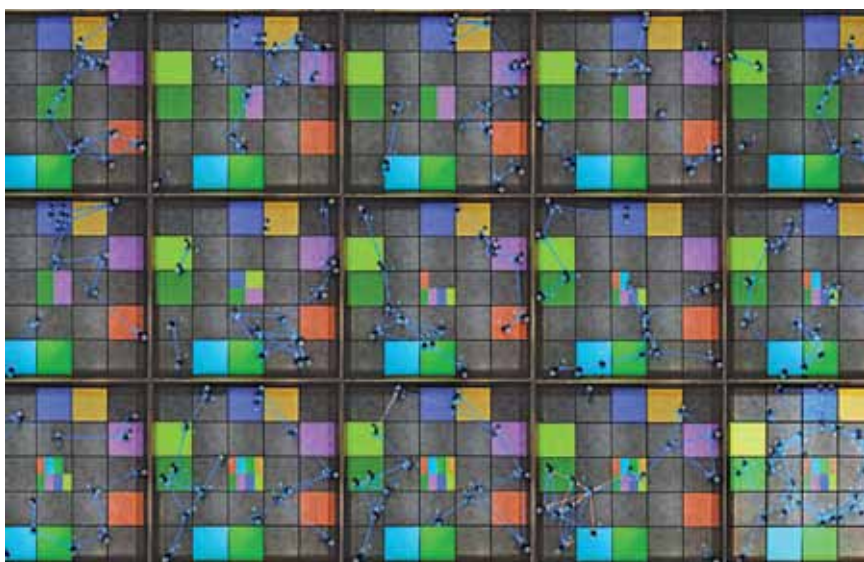


A Cornell-led collaboration created a system that uses combustion to inflate silicone membrane 'dots,' which could someday serve as a dynamic braille display for electronics. The stretchable technology could also be incorporated into soft robots, surgical tools, and wearable virtual reality equipment (Credit: <https://news.cornell.edu>)

Imagine an iPad or a Kindle for the blind, with inflatable braille that changes shape under a user's touch. A collaboration between researchers from Cornell University and Technion-Israel Institute of Technology has made a crucial component for such a technology: a haptic array of densely packed actuators that cause silicone membrane 'dots' to pop up when triggered by combustion. The system comprises moulded silicone and microfluidic liquid metal traces, in which liquid metal electrodes cause a spark to ignite a microscale volume of premixed methane and oxygen. The technology is also stretchable and conformable, and could be incorporated into a range of applications such as soft robots, wearable virtual reality equipment, or surgical tools.

Secure communication for robots with blockchain

Blockchain is typically used as a secure ledger for cryptocurrencies where each block contains information it is meant to store and a 'hash' of the information in the previous block in the chain. A blockchain offers a tamper-proof record of all transactions. Now, according to a study by researchers at MIT and Polytechnic University of Madrid, blockchain technology can also be used as a communication tool for a team of robots to provide security and safeguard against deception. So, the information stored in each block is a set of directions from a leader robot to followers. If a malicious robot attempts to alter the content of a block, it will change the block hash. Thus, the altered block will no longer be connected to the chain, which could be easily ignored by follower robots.



A team of robots collaborating to search for and then retrieve lost objects. The use of blockchain technology could enable secure, tamper-proof communication among the robots as they complete their task (Credit: <https://news.mit.edu>)